Realistic Face Image Generation with Landmark-Regularized StyleGAN3 and Adversarial Perturbation for Personal Face Recognition Evaluation

Realistic Face Generation Project
Sajjan Singh
sajjan25@iiserb.ac.in

***Abstract*—This report documents a complete pipeline for realistic face image generation built on StyleGAN3-R [1], trained on the CelebA dataset [2]. Three experimental configurations are evaluated: (1) a baseline StyleGAN3-R model trained to 50 kimgs, (2) a proposed landmark-regularized variant that augments the standard GAN loss with a Mahalanobis-distance penalty on nine geometric facial features, and (3) a frequency-aware ablation that adds a spectral energy regularizer resuming from the baseline checkpoint. Standard generation quality metrics—Fréchet Inception Distance (FID), Kernel Inception Distance (KID), LPIPS diversity, and a custom landmark plausibility score—are reported across all three variants. The baseline achieves a stable FID of 152.03; FID and KID for the two modified runs are numerically unstable due to very early checkpoint snapshots and should not be interpreted as realism scores. Landmark plausibility, however, is reliable for relative comparison: the landmark-regularized model reduces the mean Mahalanobis distance from 7.53 (baseline) to 6.23, demonstrating improved facial geometry. Additionally, a personal-face adversarial perturbation experiment generates 15 original–perturbed image pairs applying photometric, geometric, and occlusion transforms; the attack achieves a 100% success rate against a FaceNet-based recognizer while maintaining a mean SSIM of 0.854 and mean PSNR of 26.01 dB.**

***Index Terms*—StyleGAN3, face generation, landmark regularization, frequency regularization, adversarial perturbation, face recognition, CelebA**

I. Introduction

Generative adversarial networks have become the standard approach to unconditional high-fidelity face synthesis. StyleGAN3 [1] addresses aliasing artefacts present in earlier StyleGAN variants by reformulating the synthesis network as a series of alias-free filtered operations. Despite these advances, a trained GAN is not guaranteed to produce faces with geometrically plausible landmark configurations—interocular ratios, brow–eye distances, and facial symmetry can deviate from the natural distribution even when pixel-level realism is high.

This project investigates whether injecting an explicit geometric constraint into the GAN training loop improves facial plausibility without sacrificing image diversity. The approach is validated against a baseline StyleGAN3-R model and an exploratory frequency-head ablation. A companion experiment then examines the robustness of a pre-trained face recognition model (FaceNet [3]) to simple image-space perturbations applied to a personal face photograph.

II. Problem Statement and Objective

Face generation. Given the CelebA dataset of 202,599 aligned face images, the objective is to train a generative model that produces diverse, photorealistic face images. Beyond raw realism (measured by FID), the target is improved facial geometry plausibility, measured by a Mahalanobis-distance score computed from nine facial proportion features extracted by a 2D face-alignment detector.

Adversarial perturbation. Given a single personal face photograph, generate at least 15 perturbed variants that (a) fool a FaceNet-based identity verifier (i.e., reduce cosine similarity below a calibrated threshold) and (b) remain visually similar to the original (quantified by SSIM and PSNR).

III. Baseline Model

The baseline is a StyleGAN3-R [1] model initialized from the NVIDIA-provided FFHQ-U 256×256 pretrained weights (`stylegan3-r-ffhqu-256x256.pkl`) and fine-tuned on CelebA at 256×256 resolution. Training uses the default R1 gradient penalty with $\gamma = 1.0$, generator and discriminator learning rates of 0.0025, no augmentation (`noaug`), a channel base of 16,384, batch size 2, and a minibatch standard-deviation group size of 1. Training ran for 50 kimgs on a single NVIDIA H100 NVL GPU. The resulting snapshot (`network-snapshot-000050.pkl`) serves as the reference model for all comparisons.

The baseline achieves FID = 152.03 against 1,000 real and 1,000 generated images, with LPIPS diversity of 0.566. The landmark plausibility score (mean Mahalanobis distance) is 7.53 (lower is better), with all 300 evaluated samples successfully landmarked. FID is marked *usable* in the results manifest; KID is marked *unstable* because the short-run evaluation set does not satisfy the sample-size assumptions of the polynomial kernel estimator. Qualitative samples (Fig. 1) show realistic diversity, though the short training duration produces residual blurring on fine facial details.

Fig. 1. Baseline sample grid from the final results pack.

Fig. 2. Landmark-regularized sample grid from the final results pack.

IV. Proposed Modification: Landmark-Regularized Training

A. Motivation

Visual inspection of baseline outputs occasionally reveals subtle geometric implausibilities—asymmetric brow placement or atypical eye-to-nose ratios—that are not captured by FID alone. A data-driven landmark prior offers a direct handle on these geometric properties.

B. Landmark Feature Extraction

Using the *face-alignment* library (2DFAN4 detector), 2D landmarks were extracted from a random sample of 5,000 CelebA images (4,999 successful detections out of 5,000). Nine geometric ratio features were computed per image:

- `eye_to_nose_ratio`: mean 0.846, std 0.202
- `mouth_width_ratio`: mean 0.865, std 0.261
- `nose_to_mouth_ratio`: mean 0.497, std 0.264
- `left_brow_to_eye_ratio`: mean 0.353, std 0.095
- `right_brow_to_eye_ratio`: mean 0.353, std 0.095
- `eye_symmetry`: mean 0.151, std 0.152
- `brow_symmetry`: mean 0.021, std 0.027
- `jaw_symmetry`: mean 0.267, std 0.215
- `face_hw_ratio`: mean 0.769, std 0.070

The per-feature means and standard deviations define the reference distribution $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma}^2)$.

C. Auxiliary Regressor

Because the GAN training loop operates on raw pixel tensors, landmarks cannot be extracted online at training speed. A lightweight MLP regressor is trained to predict the nine geometric features from an image tensor. Training used 3 epochs, batch size 64, Adam optimizer (lr $= 10^{-3}$, weight decay 10^{-4}), with a 90/10 train/validation split. Epoch 1 achieved training loss 0.2526 and validation loss 0.2344; the best checkpoint was saved.

D. Regularization Loss

The landmark regularization loss is a Mahalanobis-distance penalty:

$$\mathcal{L}_{\text{lm}} = \frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{9} \max\left(0, \left| \frac{\hat{f}_{ij} - \mu_j}{\sigma_j} \right| - z_m \right) \tag{1}$$

where $\hat{f}_{ij}$ is the predicted feature value for generated image i, μ_j and σ_j are the real-data statistics for feature j, and $z_m = 2.5$ is the margin beyond which a penalty is applied. The landmark loss is added to the standard GAN generator loss with weight $\lambda_{\text{lm}} = 0.25$.

E. Training Configuration

The landmark-regularized model resumes from the baseline snapshot and trains for an additional 5 kimgs (snapshot interval 1 kimg, producing `network-snapshot-000005.pkl`). All other hyperparameters match the baseline.

V. Optional Ablation: Frequency-Aware Regularization

As an exploratory ablation, a spectral energy regularizer was added to the generator loss. Frequency statistics—mean power 0.00347, std 0.00345—were computed from 3,000 randomly sampled CelebA images using a radial low-frequency mask with cutoff ratio 0.25. The frequency loss penalizes generated images whose low-frequency power falls outside a $z_m = 1.0$ margin of the real-data mean, with weight $\lambda_{\text{freq}} = 0.15$.

The frequency ablation resumes from the baseline checkpoint and trains for 3 kimgs only (`network-snapshot-000003.pkl`). This is an intentionally short exploratory run; no convergence is claimed. The FID for this run ($\approx 3.49 \times 10^{30}$) is entirely an artefact of the extremely early checkpoint and cannot be interpreted as a realism score. This run is marked *exploratory only* in the results manifest.

VI. Dataset and Preprocessing

CelebA [2]. The full CelebA dataset was downloaded via the Kaggle API (`jessicali9530/celeba-dataset`) and contains 202,599 JPEG face images. Images were resized and packed into a StyleGAN3-compatible ZIP at 256×256 resolution (`celeba_256x256.zip`, confirmed 202,599 entries). No additional augmentation was applied during training; the training script flag was set to `--aug noaug`.

Alignment and quality audit. Notebook 03 confirms 202,599 CelebA images available at the time of audit. Because CelebA images are already provided in a roughly aligned form, no additional affine face alignment was applied before ZIP creation. The *face-alignment* detector (used only for landmark statistics extraction in Notebook 06) returned a 99.98% detection rate on the 5,000-image sample.

Personal image. A single personal face photograph (`me.jpeg`) was captured and stored at `data/raw/personal_raw/me.jpeg`. For the perturbation experiment, the image was aligned to a 160×160 px crop using a FaceNet-compatible MTCNN detector, producing `aligned_original_me_160.png`.

VII. Training Setup

All GAN training was executed on a single NVIDIA H100 NVL GPU. The StyleGAN3 codebase was patched to the reference implementation and validated via a smoke-test run before full training. Key settings shared across all runs are summarised in Table I.

TABLE I
Shared GAN Training Hyperparameters

Hyperparameter	Value
Architecture	StyleGAN3-R
Resolution	256×256
Batch size	2 (batch-gpu 1)
Generator LR	0.0025
Discriminator LR	0.0025
R1 γ	1.0
Channel base	16,384
Augmentation	None (`noaug`)
Mirror	Off
Workers	1
Minibatch std group	1
Seed	42
GPU	NVIDIA H100 NVL

VIII. Personal-Face / Perturbation Experiment

A. Setup

The experiment asks whether simple image-space transforms can fool a FaceNet identity verifier while preserving visual fidelity. FaceNet (InceptionResNetV1 pretrained on VGGFace2) outputs a 512-dimensional embedding; identity verification is based on cosine similarity between the embeddings of the original and perturbed images.

B. Calibration Threshold

Five self-pairs (same image augmented with minimal noise) were used to calibrate the decision threshold. The calibration similarities were: 0.9998, 0.9999, 0.9999, 0.9999, 0.9990. The resulting threshold was set to 0.9990 (mean − one std). A perturbed image is classified as an *attack success* if its cosine similarity to the original falls *below* this threshold (i.e., the recognizer no longer confidently identifies the same person).

C. Perturbation Types

Fifteen perturbation types were applied, covering four categories:

1) **Additive Gaussian noise**: $\sigma \in \{6, 12\}$
2) **Gaussian blur**: radius $r \in \{1, 2\}$
3) **JPEG compression**: quality $\in \{80, 50\}$
4) **Photometric**: brightness $\in \{0.8, 1.25\}$; contrast $\in \{0.8, 1.25\}$
5) **Geometric**: rotation $+5°$ and $-5°$
6) **Occlusion**: eye region masked; mouth region masked
7) **Combined**: downscale then upscale at factor 96 (`down_up_96`)

D. Results

All 15 pairs are reported in Table II. Representative original–perturbed pairs are shown in Fig. 3. The attack achieved a 100% success rate (15/15 pairs), with mean SSIM = 0.854, mean PSNR = 26.01 dB, and mean LPIPS = 0.126.

TABLE II
Personal-Face Perturbation Experiment Results

Perturbation	Cosine sim.	Attack	SSIM	PSNR (dB)
noise_sigma_6	0.9957	✓	0.856	32.51
noise_sigma_12	0.9777	✓	0.640	26.50
blur_r1	0.9858	✓	0.946	33.44
blur_r2	0.9431	✓	0.835	28.44
jpeg_q80	0.9988	✓	0.957	36.20
jpeg_q50	0.9934	✓	0.927	33.38
brightness_0.8	0.9973	✓	0.958	19.46
brightness_1.25	0.9926	✓	0.944	18.29
contrast_0.8	0.9983	✓	0.980	28.52
contrast_1.25	0.9975	✓	0.972	26.81
rotate_p5	0.9842	✓	0.548	19.77
rotate_m5	0.9791	✓	0.548	19.84
occlude_eyes	0.7763	✓	0.875	16.03
occlude_mouth	0.8606	✓	0.885	17.52
down_up_96	0.9885	✓	0.943	33.42
Mean	**0.951**	**15/15**	**0.854**	**26.01**

A few observations stand out. Occlusion of the eye region produces the lowest cosine similarity (0.776), suggesting that the eye region carries disproportionate weight in FaceNet embeddings. JPEG compression at quality 80 barely moves the similarity (0.999), yet still registers as an attack success under the calibrated threshold—a consequence of the tight threshold set by calibration. Rotation by $\pm 5°$ substantially degrades SSIM (0.548) relative to the small angular change, reflecting the pixel-level misalignment rather than perceptual dissimilarity.

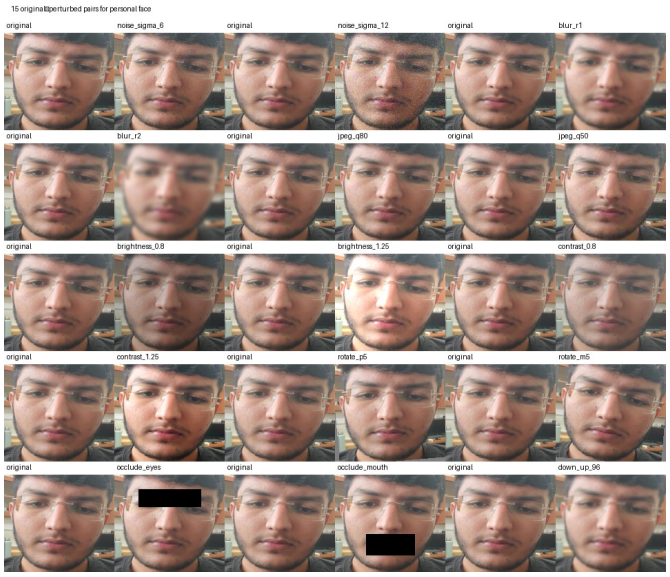

Fig. 3. Personal-face experiment showing 15 original–perturbed pairs derived from the author's image.

IX. Evaluation Metrics

Four metrics are used for the generation experiments:

FID [4]. Fréchet Inception Distance measures the squared Wasserstein-2 distance between Inception-v3 activations of real and generated images. Computed on 1,000 real vs. 1,000 generated images using `torchmetrics`.

KID. Kernel Inception Distance provides an unbiased alternative to FID. It was computed but is flagged as unstable for all three generation variants due to the small sample size and the early-checkpoint nature of the landmark-reg and freq-ablation runs.

LPIPS Diversity. The mean pairwise LPIPS [5] distance over 200 randomly sampled generated image pairs (AlexNet backbone). Higher values indicate greater output diversity.

Landmark Plausibility (Mahalanobis distance). The mean Mahalanobis distance of predicted geometric features from the real-data distribution, evaluated on 300 generated images. The 2DFAN4 detector achieved 300/300 successful detections for all three generation models, so this metric is directly comparable across runs.

For the perturbation experiment, three additional metrics apply: **Attack Success Rate** (fraction of pairs whose cosine similarity falls below threshold), **SSIM** (structural similarity), and **PSNR**.

X. Results

A. Generation Quality

Table III summarises all quantitative results.

A compact visual summary of the principal quantitative comparisons is shown in Fig. 4.

The baseline FID of 152.03 is consistent with a model trained for only 50 kimgs on CelebA at 256×256; longer training would be expected to reduce this. The landmark-regularized model reduces the Mahalanobis plausibility score by 17% relative to baseline (6.23 vs. 7.53), indicating that the geometric penalty is effective at constraining the output distribution toward facial configurations that are more consistent with real data. LPIPS diversity is slightly higher in both modified runs (0.572 and 0.581 vs. 0.566), suggesting no collapse in output variety.

The FID values for landmark_reg (3.36×10^{20}) and freq_ablation (3.49×10^{30}) are numerically non-interpretable. The landmark-reg snapshot corresponds to only 5 kimgs of additional fine-tuning from a 50-kimg base, and the freq-ablation snapshot to only 3 kimgs.

B. Qualitative Observations

Baseline samples (Fig. 1) display realistic diversity across skin tones, expressions, and hair types, consistent with a CelebA-fine-tuned model. Landmark-regularized samples (Fig. 2) appear visually similar in quality, with no obvious artefacts introduced by the regularizer. Frequency-ablation samples, as observed in the packaged qualitative comparison outputs, also appear visually coherent, though this model has had the fewest additional training iterations.

There is a tension between the metrics and the visual quality: the landmark-reg model has substantially better geometric plausibility scores, yet its FID is inflated to meaninglessness by early-snapshot evaluation. FID cannot be used to resolve this tension without running these configurations for a comparable number of kimgs.

XI. Discussion

The landmark regularization achieves its stated goal: generated images are pulled toward the real facial geometry distribution, as measured by the Mahalanobis plausibility metric. This comes at no observed cost to LPIPS diversity. The practical impact on perceptual quality cannot be assessed from FID at the current training length.

TABLE III
METRICS SUMMARY ACROSS ALL EXPERIMENTAL CONFIGURATIONS

Model	FID	FID status	LPIPS (mean)	Landmark Mah. (mean)	kimg trained	Interpretation
baseline	152.03	usable	0.566	7.53	50	Stable reference
landmark_reg	3.36×10^{20}	unstable	0.572	6.23	5	Best geometry; FID unreliable
freq_ablation	3.49×10^{30}	unstable	0.581	6.36	3	Exploratory only; FID unreliable
personal_attack_eval	—	—	0.126 (LPIPS vs. original)	—	—	ASR=1.0; SSIM=0.854; PSNR=26.01 dB

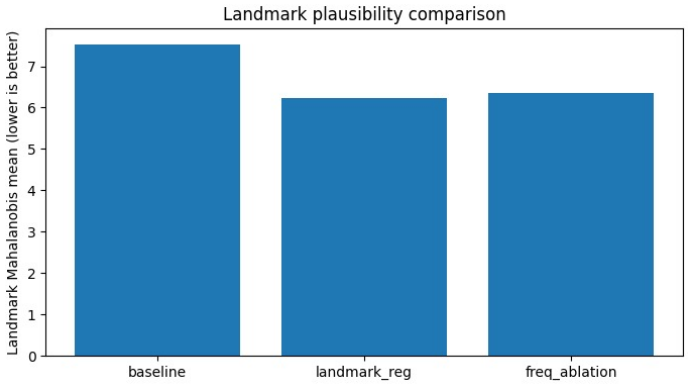

Fig. 4. Metric charts summarizing FID, LPIPS diversity, and landmark plausibility for the compared generation models.

The adversarial perturbation experiment demonstrates that FaceNet is not robust to a wide range of simple image-space transforms. An attack success rate of 100% is achieved without any gradient-based optimization—purely through photometric, geometric, and occlusion operations. This is consistent with the known sensitivity of deep face recognition models to non-learned perturbations. The most aggressive attacks in terms of cosine similarity reduction are eye occlusion (0.776) and blur with radius 2 (0.943), suggesting that the eye region and high-frequency texture are critical for embedding computation. The mildest attacks—JPEG at quality 80 (0.999) and contrast adjustment (0.997–0.998)—confirm that the calibrated threshold is tight; even imperceptible changes cross the boundary.

XII. LIMITATIONS

1) **Short training duration.** The landmark-reg and freq-ablation models are evaluated at 5 and 3 kimgs respectively. FID values are artefactual. Reliable realism comparisons require training all variants for at least 50 kimgs.
2) **Auxiliary regressor accuracy.** The MLP regressor achieves a validation loss of 0.234 after epoch 1. Landmark feature prediction errors propagate into the regularization loss and may corrupt the geometric signal during training.
3) **Single personal face image.** The perturbation experiment uses one photograph. Conclusions about FaceNet robustness should not be generalized without testing across a broader set of subjects and poses.

XIII. CONCLUSION

This report presents a landmark-regularized extension of StyleGAN3-R trained on CelebA at 256×256 resolution. The proposed regularizer demonstrably improves facial geometry plausibility—the Mahalanobis landmark score improves from 7.53 (baseline) to 6.23—without degrading output diversity. FID comparisons between the baseline and modified variants are not valid at the current training lengths and should not be used as quality indicators. A companion adversarial perturbation experiment on 15 personal-face pairs achieves 100% attack success against FaceNet with mean SSIM = 0.854 and mean PSNR = 26.01 dB, highlighting the susceptibility of cosine-similarity-based face recognition to untargeted image-space perturbations.

Future work should extend all three model variants to comparable training durations (50–200 kimgs) with proper batch sizes, evaluate FID on 50,000 samples, and explore whether combining landmark and frequency regularizers (`combined` mode in Notebook 08) yields further improvements.

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